

A brief write-up on 'SmartKheti'

SmartKheti is an Innovative solution that uses latest technology called 'oneM2M' to provide farmers with a very useful tool to improve farm yield while reducing the input costs on water, fertilisers, pesticides, labour etc. It provides new job opportunities in the villages and attracts youth to the villages.

The Solution uses Solar powered sensors and aggregators to collect and transport soil data to the servers using WiFi or Cellular (2G/4G/5G) network. The solutions has robust hardware and software that are already being used by various agencies at various locations including ICAR Andaman Nicobar Islands, CDoT Bangalore, Pusa Agriculture Research centre, village near Nagpur in Maharashtra. It includes wireless soil sensors that are buried in the farm and are connected to the Gateway wirelessly. SmartKheti Gateway collects the soil parameters and communicates to the Central server. The central server processes the received data to generate dashboards used for various purposes. The farmers get a free application on their phone to have a real time information of their crops, they can automate the irrigation and get a number of other value added services. Farmers can book 'online' rental of the expensive farming equipment, buy seeds, fertilizers, pesticides and also have access to markets for selling their produce.

It also provides a dashboard for the Govt agencies to monitor farm data of a particular Block / District / State. Government agencies can assess expected produce to plan better schemes for farmers. Government agencies can educate the farmers and issue advance alerts about weather or spreading of Crop deceases.

This product was designed after multiple iterations with farmers from all over India 'Kisan Melas'. As per the evaluation by the experts at ICAR this tool by using AI and IoT can help farmers to increase their yield by 15% and reduce water consumption by 30% and labour by 50%.

A farmer when registered on SmartKheti platform will get a specific 'Farm ID' which will carry the farmers Aadhar no., Bank details, Farm visibility on digital map. In the case of flood or drought or any other natural calamities the farm ID of affected farmers will be identified from the digital map to estimate the extent of losses and in case Govt. plans to compensate directly to the farmers it can be passed on to the affected farmers. Similarly for crop insurance this information will be very useful for the insurance companies.

Following are the advantages of SmartKheti solution:

1. Increase crop yields and production efficiency using technology. As per the estimate by the experts it saves 30% water requirements.
2. Reduce the environmental impact of agriculture through sustainable practices.
3. Provide real-time monitoring of crops and soil conditions to optimize irrigation and fertilization.
4. Improve the quality and safety of crops by detecting and preventing diseases and pests.
5. Enable farmers to make data-driven decisions using analytics and predictive modelling.
6. To reduce labour costs by automating routine tasks such as planting, harvesting, and monitoring. As per the estimate by expert it saves 50% of manpower.
7. To improve food security by increasing the availability and accessibility of locally grown produce.
8. To promote precision agriculture practices that minimize waste and maximize productivity.
9. To empower farmers with tools and knowledge to manage their operations more effectively.
10. To create a more resilient and sustainable agricultural sector that can adapt to changing conditions and challenges.
11. Precision farming: Using sensors and data analytics to monitor soil conditions, weather patterns, and crop growth to optimize irrigation, fertilization, and pest control.
12. Rental of Agricultural drones: Farmers can rent drones to monitor crop health, detect crop diseases and pests, and perform tasks such as spraying pesticides and fertilizers.
13. Farm management software: Using software tools to manage farm operations, including crop planning, field mapping, and data analysis.